Pharmalink

B2B Medicine Distribution Platform

Case Study

For ML Beginners: Classification · Regression · Bagging · Boosting

Problem Identification

TYPE OF LEARNING

Supervised Learning

The target variable will_churn_next_30d is already labeled (1 = Churn, 0 = Active). The model learns from these historical labeled examples to predict future pharmacy behavior.

TYPE OF PROBLEM

Binary Classification

The output is one of two categories — churn or stay active. We are not predicting a number, so Regression does not apply. Each pharmacy gets assigned to one of two classes.

ML CATEGORY

Churn Prediction

This is a Churn Prediction task. Best solved using Boosting (XGBoost) or Bagging (Random Forest) as the dataset has 42 features and 1.5M rows — ensemble methods handle this well.

The 5-Stage ML Pipeline

01

Data Collection

Pull 15 months of records: last_order_date, order_frequency_60d, credit_utilization_ratio, payment_delay_days, city_tier, store_type.

02

Data Preprocessing

Fix nulls in app_sessions_30d. Normalize avg_order_value. Create days_since_last_order from last_order_date.

03

Model Training

Train XGBoost or Random Forest on will_churn_next_30d. Include city_tier as a feature. Split into train/test before fitting.

04

Model Evaluation

Avoid accuracy alone — dataset is imbalanced. Evaluate using Recall, Precision, and AUC-ROC. Recall on churn class is most critical.

05

Deployment

Run daily batch scoring. Output ranked list by churn probability × avg_order_value for sales team prioritization.

Evaluation Strategy

> *"We can afford 500 unnecessary calls. But losing a ₹8 lakh/month hospital account without warning — that's unacceptable."*
>
> — Rohan Mehta, CRO

✓ PRIORITIZE: RECALL

Goal: Catch every pharmacy about to churn
Risk accepted: Wasting some calls on loyal pharmacies
False Negatives: Predicted active → Actually churned → Revenue lost
CRO stance: High Recall is non-negotiable

✗ NOT PRIORITIZED: PRECISION

Goal: Only flag pharmacies truly about to churn
Risk accepted: Silently missing real churners
False Positives: Predicted churn → Actually fine → Wasted call
CRO stance: 500 wasted calls are affordable

Handling Tier-Based Behavior

⚠️ Core Problem: A Tier-3 pharmacy ordering once every 45 days is healthy. The 30-day churn rule wrongly flags it. A global model learns the wrong pattern.

Tier-1 Separate Model

30-day inactivity threshold

Key Features

```
app_sessions_30d
order_frequency_60d
```

Frequent, smaller orders. 30-day silence is a real red flag.

Tier-2 Separate Model

45-day inactivity threshold

Key Features

```
avg_order_value
credit_utilization_ratio
```

Moderate bulk orders. Slightly longer window needed for fair assessment.

Tier-3 Separate Model

60-day inactivity threshold

Key Features

```
payment_delay_days
product_diversity_score
```

Large, infrequent orders. Standard 30-day rule causes false positives.

Interaction Features + Key Takeaway

INTERACTION FEATURES TO ENGINEER

`city_tier × days_since_last_order`

40 days of inactivity means very different things for Tier-1 vs Tier-3 pharmacies.

`city_tier × order_frequency_60d`

Low frequency is normal for Tier-3 but alarming for Tier-1. This feature teaches the model that difference.

`avg_order_value × credit_utilization_ratio`

Flags financial stress in high-value accounts — the exact risk the CRO is worried about.

Bottom Line

Pharmalink's churn problem is not one problem — it's three distinct behavioral problems wearing the same label. Solve it with tier-specific Classification models (Boosting or Bagging), evaluated on Recall, deployed with revenue-weighted prioritization.